Dashiell Kosaka

Email: dashkosaka@gmail.com
Mobile: +1 (312) 203-7512
Website: dashkosaka.github.io
LinkedIn: linkedin.com/in/dash-kosaka
Residence: Austin,Texas

Skills

- **ML/DL/CV**: PyTorch, Cupy, Keras, TensorFlow, Scikit-learn, Scipy, Numpy, Pandas, RAG, LLM, RNN, GAN, CNN, Reinforcement Learning, Deep Q Learning, Clustering, Segmentation, Detection, Bayesian Modeling, OpenCV, SIFT, RANSAC, OCR, Regression, Classification
- **Robotics**: ROS, Embedded Systems, Microcontrollers, Multi-Object Tracking, Kalman Filtering, Particle Filtering, SLAM, Data Association, LAP, Occupancy Grid Mapping, Radar, Camera, Sensor Fusion, Perception
- **Cloud & DevOps**: Agile, Bitbucket, Confluence, Jira, Git, Mercurial, AWS, GCP, Docker
- **Programming**: Python, C, C++, WSL, MATLAB, CUDA, x86, RISC-V, Bash, JavaScript, HTML, SystemVerilog
- **Other**: Data Analysis, Data Visualization, Scripting & Automation, Prototyping, Signal Processing, Remote Sensing
- **Languages**: English (Native), Japanese (Conversational), Spanish (Basic)

Work Experience

Senior Systems Engineer – Algorithm & Radar Feature Development
Uhnder Inc. – Austin, TX | January 2023 – Present

- **Prototyped** Python and MATLAB solutions for 5 radar interference management algorithms, boosting interference resilience by over 10x and restoring more than 20 dB of target SNR
- **Integrated** interference management algorithms into a real-time C++ embedded radar system, meeting strict 20 Hz operational requirements under memory constraints
- **Architected** an analytics framework to identify edge cases across 8 production radar models, improving stability against millions of radar sensors in use today
- **Automated** system-level testing for interference sensing and mitigation APIs, accelerating deployment cycles and enhancing testing efficiency by 500% through Python, Bash, and Bitbucket pipelines
- **Presented** radar feature developments to potential investors, securing multimillion-dollar investments by showcasing algorithmic improvements

Systems Engineer – Algorithm & Radar Feature Development
Uhnder Inc. – Austin, TX | June 2020 – December 2022

- **Deployed** and validated 4 radar perception algorithms across teams, ensuring seamless integration into our ROS stack to enhance system performance and reliability
- **Implemented** a scalable multi-object radar tracking system, delivering a C++ API for production use on tens of thousands of radar modules
- **Modeled** a Bayesian mapping algorithm for sensing static and dynamic targets, further enhancing runtime performance by 90% through CUDA and architectural optimizations

Academic Research Intern – Remote Sensing & ML/CV
National Center for Supercomputing Applications – Champaign, IL | October 2019 – June 2020

- **Engineered** a U-Net model in a deep learning pipeline for detecting waterlogged farmland, attaining 95% segmentation accuracy on 3-meter resolution PlanetLab CubeSat imagery
- **Expanded** the waterlogging detection solution from Champaign County to hundreds of counties across Illinois, processing large datasets with multi-channel GIS data
- **Resolved** extrinsic calibration errors in satellite imagery, achieving sub-meter accuracy using SIFT, RANSAC, and geometric correction techniques

Software Engineering Contractor – Embedded Software & Front-End Development
Cohesive Manufacturing – Champaign, IL | September 2019 – March 2020

- **Led** the development of an embedded Arduino system for sensor data logging, creating drivers for 3 mission-critical sensors and optimizing cloud logging with sub-20 ms latency for real-time monitoring
- **Enhanced** customer website engagement by implementing multi-language support, boosting usability and increasing global reach for manufacturing partners

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Projects

Uhnder Interference Manager
Uhnder Inc. | Nov 2021 – Present
- **Investigated** and deployed signal processing algorithms to detect, classify, and mitigate interference
- **Evaluated** algorithms on an embedded radar system, establishing KPIs to measure performance improvements

Uhnder Object Layer
Uhnder Inc. | June 2020 – Oct 2021
- **Explored** perception-related radar algorithms to enhance output through refinement, tracking, and mapping
- **Facilitated** radar feature integration by prototyping ROS nodes and merging them into the ROS stack

OpenRadar [github.com/PreSenseRadar/OpenRadar | arxiv.org/abs/1912.12395]
PreSense | January 2019 – June 2020
- **Oversaw** the development of an open-source radar signal processing library, cited globally by researchers
- **Contributed** tools for generalized radar ADC data processing and AI-related functionalities

AlphaSmash [github.com/DashKosaka/AlphaSmash]
HackIllinois 2019 Hackathon | February 2019
- **Developed** a Double Deep Q Learning (DDQN) agent to autonomously play Super Smash Bros. Ultimate
- **Assembled** a hardware interface using an Arduino and camera to interpret state space and execute actions

Deep Fake Farms
AGCO Accelerator 2019 Hackathon | February 2019
- **Utilized** a GAN in PyTorch to synthesize agricultural imagery, enabling large-scale data augmentation
- **Created** a scalable generation method capable of producing arbitrarily sized outputs

EIE.io
AGCO Accelerator 2019 Hackathon | February 2019
- **Built** an object detection pipeline with TensorFlow and React to monitor livestock health
- **Leveraged** GCP's Vision API and Firebase Realtime Database to store data and alert users of unhealthy animals

TyperML
UIUC Personal | December 2018
- **Designed** an interactive typing game to improve users' typing skills by learning commonly missed sequences
- **Implemented** RNN models to generate text sequences while predicting typing speed and accuracy

RISC-V Architecture [github.com/DashKosaka/RISC-V]
UIUC ECE | Top 5 | October 2018 – December 2018
- **Pipelined** a 5-stage RISC-V CPU, optimizing throughput and latency while executing on an FPGA
- **Enhanced** the architecture with a PHT branch predictor and L3 cache, earning a top 5 competition finish

SrirachOS Operating System [github.com/DashKosaka/SrirachOS]
Microsoft Operating System Design Competition | Top 4 | February 2018 – May 2018
- **Engineered** the file system architecture, pagination, and context switching for a CLI-based OS
- **Upgraded** the system with a search function and command history, improving user experience and efficiency

FPGA Stepmania [github.com/DashKosaka/fpga_stepmania]
UIUC ECE | December 2017
- **Orchestrated** a state machine in SystemVerilog to manage gameplay logic for a StepMania variant
- **Devised** a VGA display driver to render game assets directly on-screen using the Altera DE2-115 FPGA

Education

B.S. Computer Engineering with Honors
University of Illinois at Urbana-Champaign | September 2015 – May 2019
- Artificial Intelligence, Deep/Machine Learning, Computer Vision, Data Structures & Algorithms
- Operating Systems, Probability & Statistics, Linear Algebra, Signal Processing, Computer Architecture